Sally Mohammadzadeh



Applied AI Engineer · Computer Engineering Student

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Profile

Applied AI engineer and co-lead of a 6-person computer vision subteam at DIANA, the Politecnico di Torino rover team (4th of 126, European Rover Challenge (ERC) 2025). Builds end-to-end pipelines in Python, from field-collected data to trained YOLOv8 models and Dockerized ETL on PostgreSQL. Computer Engineering student, graduating in September 2027.

Education

BSc in Computer Engineering, Politecnico di Torino, Turin — Sep 2027 (expected)

- Coursework: Databases, Algorithms & Data Structures, Object-Oriented Programming, Computer Architecture, Cybersecurity

Experience

Applied AI Engineer, Computer Vision Co-Lead, DIANA Student Robotics Team — Oct 2023 – Present

- 4th of 126 teams at ERC 2025: built the rover's YOLOv8 probe-detection pipeline.
- Selected YOLOv8, CVAT and OpenCV through AI-assisted research comparing detection and annotation tools.
- Ran every stage: field data collection, CVAT annotation and quality control, augmentation, training.
- Co-led the 6-person computer vision subteam, coordinating with mechanical and software subteams.
- Developed and coded the rover arm's inverse-kinematics simulation in MATLAB/Simulink, both self-taught.

Projects

SpaceX Launch Data Pipeline, personal project — 2026
github.com/sallyze/spacex-pipeline

- Engineered a Dockerized 4-stage ETL pipeline loading 205 SpaceX launches into two PostgreSQL 15 tables.
- Created a Streamlit dashboard with KPI metrics and yearly launch-success charts, queried from PostgreSQL.
- Packaged the full stack to start with one command: `docker compose up`.

Triple DES (3DES) Encryption in Pure Python, university project — 2025
github.com/sallyze/cryptography-Triple-DES

- Implemented 3DES encryption and decryption from scratch, with no external libraries.
- Coded each DES component by hand: S-boxes, Feistel network, permutations, 48-round key scheduling.
- Added ECB mode and optional PKCS-style padding, and documented the design in a technical report.

Skills

Programming: Python, C, SQL, MATLAB/Simulink
ML & Computer Vision: PyTorch, YOLOv8, OpenCV, NumPy, Pandas, CVAT
Data & Tools: PostgreSQL, Docker, Streamlit, REST APIs, Git/GitHub
Languages: Italian (A2, studying), English (C1, IELTS 7.0), Persian (native), Azerbaijani (C1)

Certifications

Building with the Claude API, Anthropic, May 2026
SpaceCert, European Space Foundation, Sep 2025
ERC Space & Robotics Industry Standard, European Rover Challenge, Sep 2024

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